PROGRAM 4: Text Translation and Paraphrasing

AIM

To perform text translation and paraphrasing using pretrained language models.

ALGORITHM

1. Import the Transformers library.
2. Load a translation model.
3. Accept text from the user.
4. Select translation or paraphrasing.
5. Generate the result.
6. Display the output.

CODE

```
from transformers import pipeline

translator = pipeline(
    "translation",
    model="Helsinki-NLP/opus-mt-en-hi"
)

paraphraser = pipeline(
    "text2text-generation",
    model="google/flan-t5-small"
)

text = input("Enter text: ")

print("1. Translation")
print("2. Paraphrasing")

choice = input("Enter choice: ")
```

```
if choice == "1":

    result = translator(text, max_length=100)
    print("\nHindi Translation:")
    print(result[0]["translation_text"])

elif choice == "2":

    prompt = "Paraphrase this sentence: " + text

    result = paraphraser(
        prompt,
        max_new_tokens=100
    )

    print("\nParaphrased Text:")
    print(result[0]["generated_text"])

else:
    print("Invalid choice")
```

INPUT

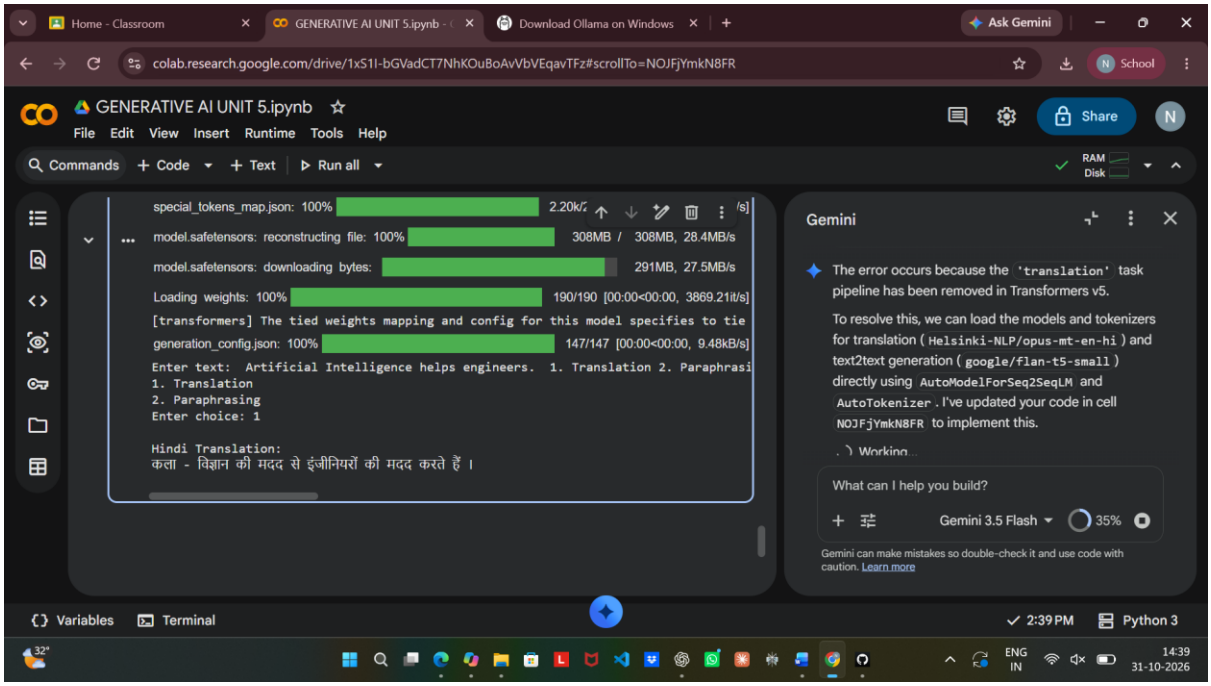
Enter text: Artificial Intelligence helps engineers.

1. Translation
2. Paraphrasing

Enter choice: 1

OUTPUT

The Hindi translation of the given sentence is displayed.



RESULT

Thus, text translation and paraphrasing were performed using pretrained language models.